Quiz 4 Solution

Question:

id	X	Y
1	1	1
2	1	0
3	0	2
4	2	4
5	3	5

Initial Centeroids

$K_1 = (1, 0.5)$ $K_2 = (1.7, 3.7)$

Solution:

1ST ITERATION

Compute Distances using Euclidian Distance → $\sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$

$d(K_1, 1) = \sqrt{(1-1)^2 + (1-0.5)^2} = 0.5$

$d(K_2, 1) = \sqrt{(1-1.7)^2 + (1-3.7)^2} = 2.789$

↳ do the same for: $d(K_1, 2)$ and $d(K_2, 2)$

$d(K_1, 3)$ and $d(K_2, 3)$

$d(K_1, 4)$ and $d(K_2, 4)$

$d(K_1, 5)$ and $d(K_2, 5)$

answers:

Computing Distances 5 marks

id	$d(K_1, id)$	$d(K_2, id)$	cluster
1	0.50	2.79	K_1
2	0.50	3.77	K_1
3	1.80	2.40	K_1
4	3.64	0.42	K_2
5	4.92	1.84	K_2

Identify clusters 2 marks { ∴ clusters → (1, 2, 3) and (4, 5)

now calculate new centeroids i.e new K_1 and K_2 using mean

Calculate New Centers 3 marks

$K_1 = \left(\frac{1+1+0}{3}, \frac{1+0+2}{3}\right)$ $K_2 = \left(\frac{2+3}{2}, \frac{4+5}{2}\right)$

$\therefore K_1 = (0.67, 1)$ $\therefore K_2 = (2.5, 4.5)$

2ND ITERATION

Compute distances using Euclidian dist and new centeroids

$d(K_1, 1) = \sqrt{(1-0.67)^2 + (1-1)^2} = 0.33$

$d(K_2, 1) = \sqrt{(1-2.5)^2 + (1-4.5)^2} = 3.81$

now do the same for rest of distances like before

Entire Second Iteration 5 marks

3 for distances

1 for clusters

1 for centeroid

result:

id	$d(K_1, id)$	$d(K_2, id)$	cluster
1	0.33	3.81	K_1
2	1.05	4.74	K_1
3	1.20	3.54	K_1
4	3.28	0.71	K_2
5	4.63	0.71	K_2

→ since clusters same as before, the centeroids will remain same i.e. $K_1(0.67, 1)$ & $K_2(2.5, 4.5)$

Total 15 Marks